

AUTONOMOUS DRONE MAPPING

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BACKGROUND

Advances in robotics and machine learning have accelerated research in autonomous aerial systems. Published work on SLAM, including ORB-SLAM3 and FAST-LIO2, represents years of doctoral research. This project distills those independent pipelines into a unified system optimizing coverage completeness, energy efficiency, and flight time.

This semester we developed a simulation-first autonomous drone pipeline spanning three integrated sub-systems: physics-based simulation, SLAM-based mapping, and reinforcement learning path planning. The simulation-first approach allows rapid iteration before hardware deployment.

Our innovation lies in combining these independently researched systems into a single cohesive pipeline capable of autonomously exploring and mapping unknown environments in real time. In doing so, we aim to produce accurate 3D digital replicas of physical spaces.

SIMULATION ENVIRONMENT

Our Environment

Isaac Sim provides a high-fidelity physics engine and photorealistic rendering for the drone. Pegasus Sim acts as the drone manager, loading the sensor-equipped UAV model and spawning the PX4 SITL (Software in the Loop) flight controller instance. Motor commands fed back from PX4 complete the closed-loop simulation.

The ROS 2 bridge converts Isaac Sim's internal sensor streams into standard ROS 2 topics consumed by downstream modules.

Trajectory commands from the RL planner return via the ROS 2 bridge to PX4 SITL.

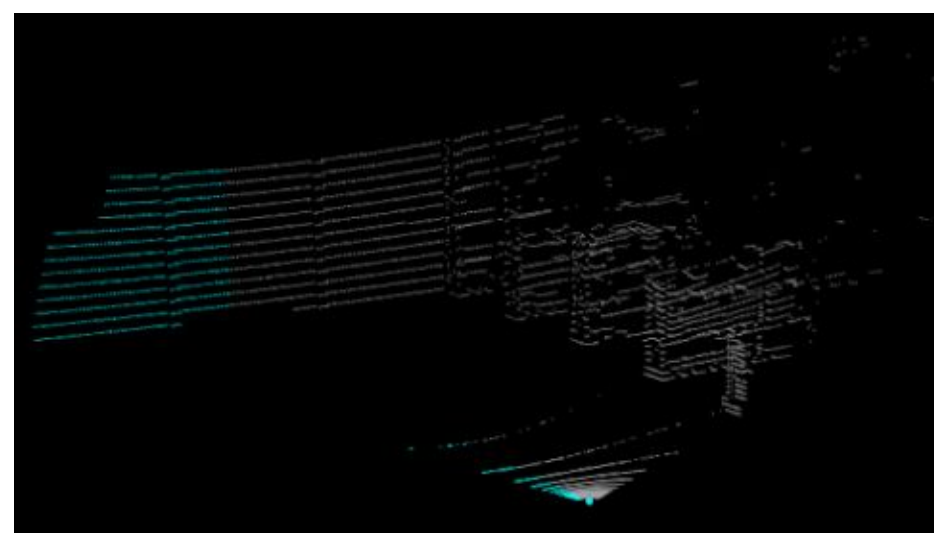


Fig 1. RViz visualization of LiDAR point cloud pointed at a series of shelves and a wall in an Isaac Sim warehouse.

Sensors

Our three main sensor topics and their associated data stream:

Stereo Camera: RGB + depth frames

LiDAR: 3D point cloud scan

IMU: linear acceleration & angular velocity

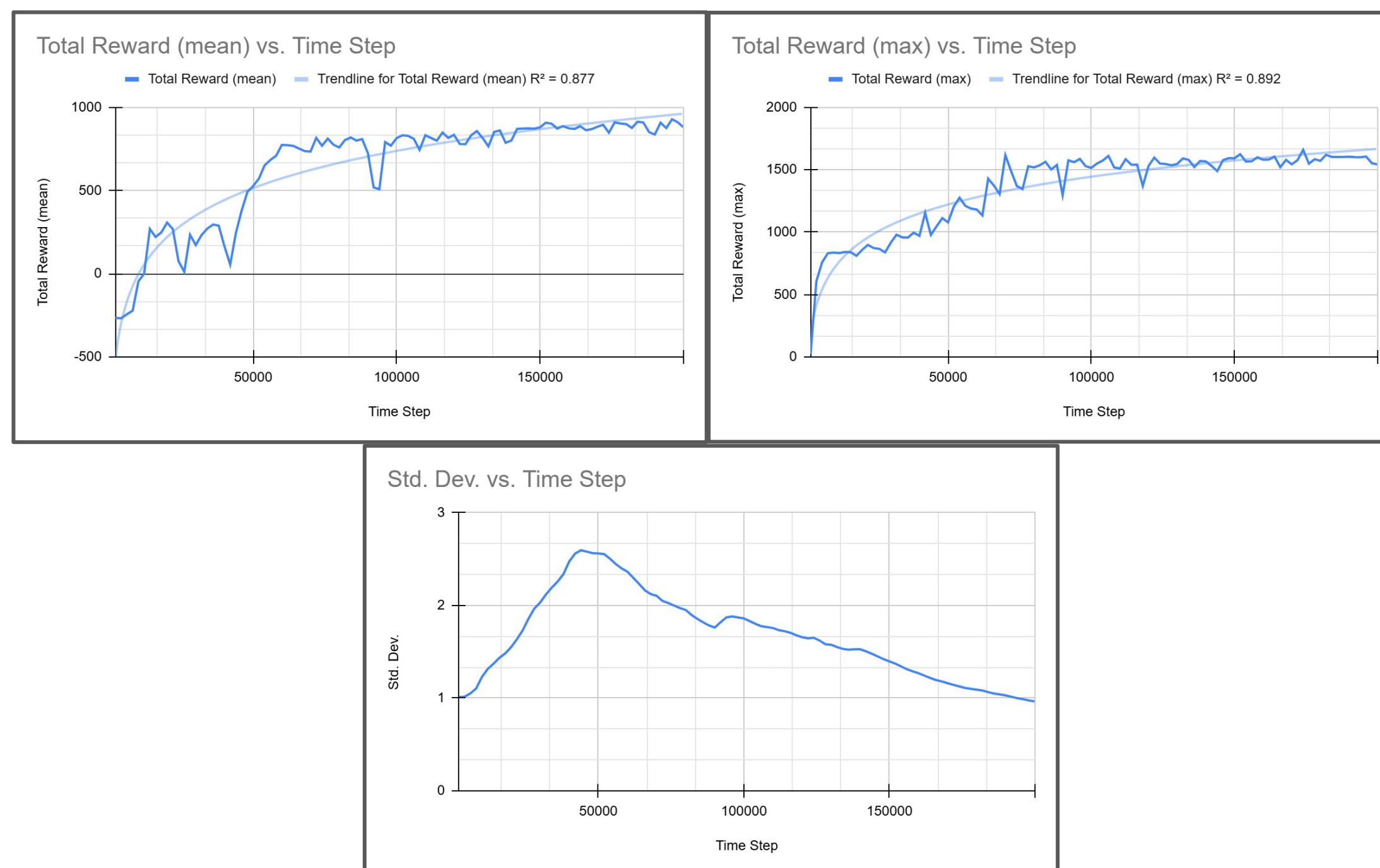
Sim-to-Real Strategy

The drone model includes a full sensor suite mirroring planned hardware. To simulate real world conditions, we plan to randomize lighting, obstacle placement, and add sensor variance in Isaac Sim.



Fig 2. Render of the quadcopter drone assembled in Fusion 360. Hardware includes Intel RealSense D455 and Hesai XT32.

RESULTS



ARCHITECTURE OVERVIEW

Simulation Layer

Isaac Sim provides the physics engine and renderer. Pegasus Sim manages the drone model and spawns a PX4 SITL flight controller instance. Motor commands from PX4 feed back into Isaac Sim, closing the control loop. The ROS 2 Bridge converts Isaac Sim's internal sensor streams into standard ROS 2 topics, including camera, LiDAR, and IMU, consumed by downstream layers.

SLAM Layer

LiDAR and IMU topics feed into Fast-LIO2, which produces accurate odometry via tightly-coupled LiDAR-inertial estimation. An Extended Kalman Filter fuses this with the flight controller's raw odometry into a single smoothed estimate. RTAB-Map ingests all original sensor data alongside the fused odometry to perform visual SLAM, loop closure, and generate a 3D occupancy grid via Octomap.

RL Planning Layer

The Octomap is passed to Frontier Detection, which segments the occupancy grid into explored and unexplored regions. The PPO policy, trained in Isaac Lab, selects the next navigation target from these frontiers. Planned trajectories are sent back to PX4 SITL via ROS 2 topics to execute the flight commands.

SIMULTANEOUS LOCALIZATION & MAPPING (SLAM)

What is SLAM?

SLAM or Simultaneous Localization & Mapping consumes raw sensor data, including LiDAR, IMU, and RGB-D information, to compute a map of drone's environment. Using this information, we are able to localize the drone by precisely predicting its position.

Role in the Pipeline

We use a version of SLAM, Visual SLAM, to build a 3D Occupancy Map (Octomap) that serves as the environment for our reinforcement learning algorithm.

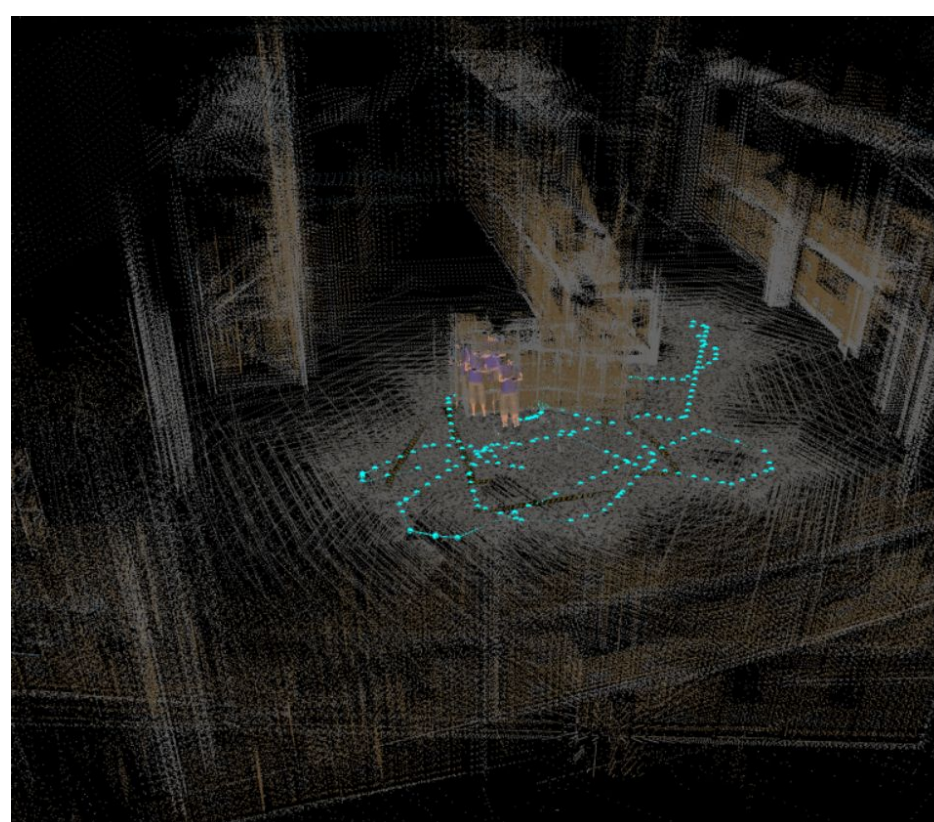


Fig. 3. RViz test of preliminary test of TurtleBot traversing warehouse with colored point cloud mappings.

How does it work?

- Raw LiDAR + IMU data are passed to Fast-LIO2 to produce highly accurate odometry estimates.
- An Extended Kalman Filter subscribes to both Fast-LIO2's precise odometry and the flight controller's raw odometry, fusing them into a single smoothed estimate.
- RTAB-Map ingests all original camera, LiDAR, and IMU data alongside the EKF's fused odometry. It stitches observations together, performs loop closure detection, and generates the full 3D Occupancy Map.

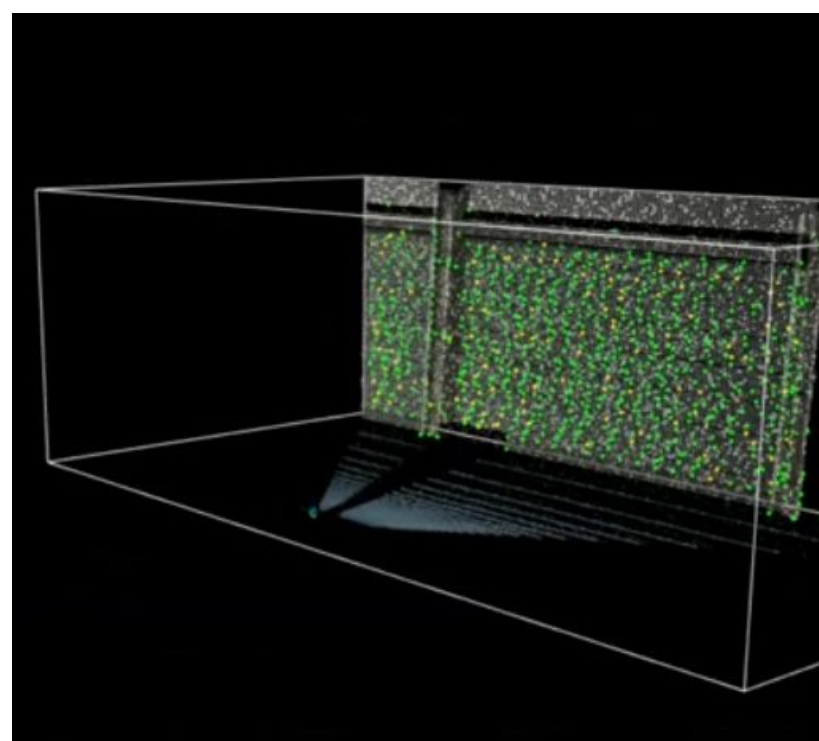


Fig. 4. RViz visualization of RTAB-Map using fused LiDAR and RGB-D data from scanning a warehouse wall in Isaac Sim.

CONCLUSION & NEXT STEPS

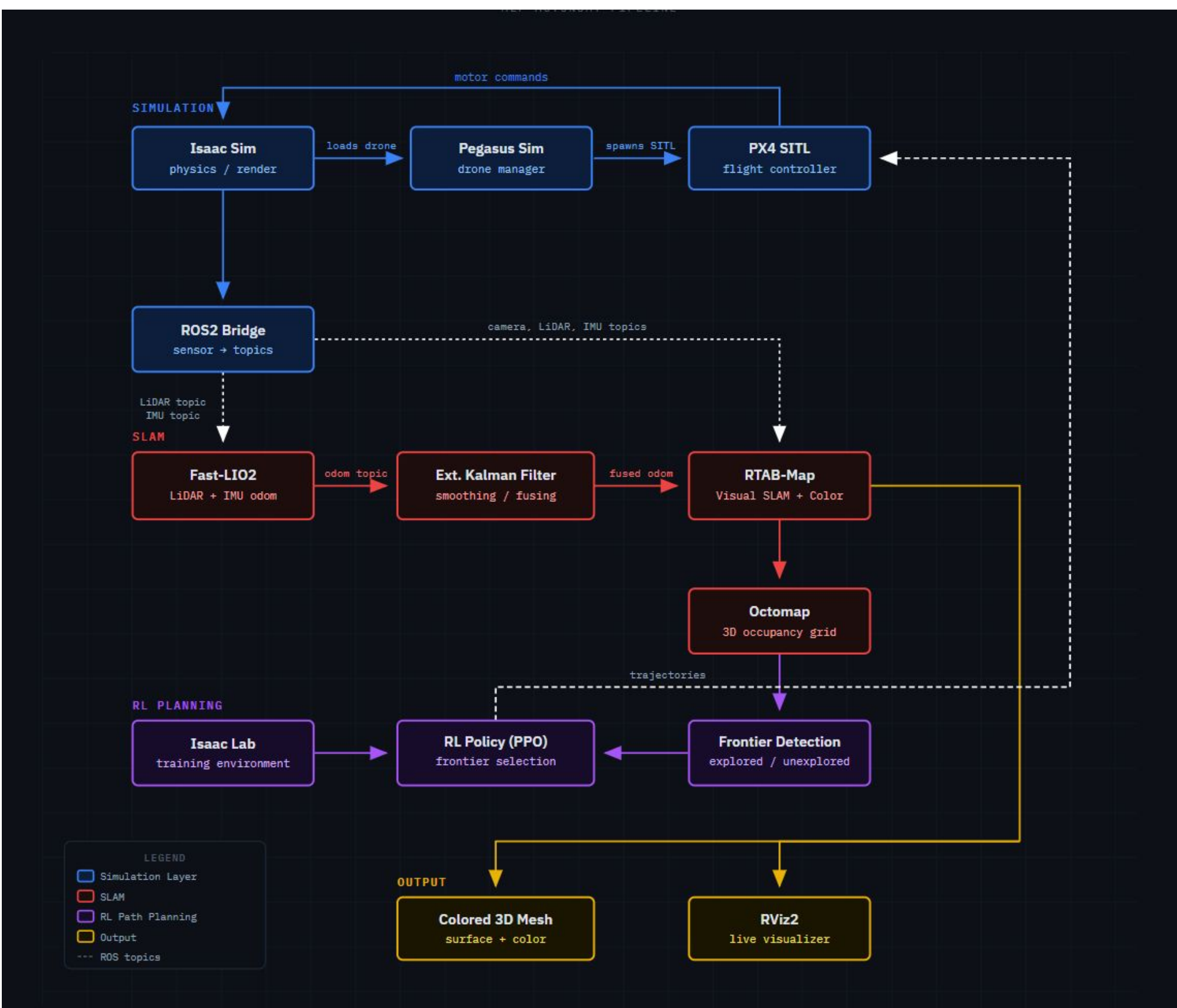
Statistical Analysis

The agent was trained using PPO for 200k timesteps in the warehouse environment. Policy standard deviation peaked early approximately 50k steps before decaying to ~1.0, reflecting a healthy exploration-to-exploitation transition as the policy converged. Mean total reward climbed from near zero to approximately 880 by ~100k steps and plateaued, indicating stable convergence. Max total reward reached ~1,570, demonstrating that the learned policy is capable of near-optimal episode performance. Together these results show the agent successfully learned a navigable policy, with the mean-max gap narrowing in the latter half of training as consistent performance improved.

Future Developments

This semester we designed and integrated a simulation-first autonomous drone mapping pipeline across three sub-systems: physics-based simulation via Isaac Sim and PX4 SITL, LiDAR-inertial SLAM using Fast-LIO2, EKF fusion, and RTAB-Map, and a PPO policy trained in Isaac Lab for coverage-maximizing path planning. Early training results show a rising reward signal, with full end-to-end pipeline integration as the immediate next milestone. The simulation-first methodology and containerized environments establish a reproducible foundation for physical hardware deployment in future semesters.

PIPELINE OVERVIEW



RL PATH PLANNING

Framework

A quadrotor learns to maximize 3D voxel coverage of a warehouse within a fixed 60-second episode. The policy is trained in Isaac Lab using skrl PPO across 2048 parallel environments on a single GPU.

State Space

State Space $S =$
{ XYZ position, orientation quaternion, linear velocity, angular velocity, LiDAR ranges, voxel coverage map, mean layer coverage fractions, normalized altitude }

Action Space

A holonomic force model with an identity quaternion applied to the orientation to simulate flight controller.

Action Space $A =$
{ Vertical Thrust (body Z force), Forward Force (body X force), Lateral force (body Y force), Yaw movement }

Note: In our initial simulations, we primarily accounted for the body XYZ forces, since in the sim-to-real transfer yaw movement is handled by the flight controller.

Reward Function Parameters

Term	Description	Max Weight
r_{coverage}	Voxel coverage gain	+20
r_{novelty}	First-visit bonus per coarse cell	+2
r_{frontier}	Velocity alignment toward unexplored frontier	+1
r_{vertical}	Amplifies coverage reward in sparse altitude layers	x0.3
r_{altitude}	+0.2 inside [1,6]m range to encourage horizontal exploration	+0.2
$r_{\text{collision}}$	0.4 m wall proximity	-5
r_{terminal}	End-of-episode coverage bonus	+25

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